σ_2-G1 at $n = 5$ via branching reduction to σ_1-G1

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Abstract

For the Hecke algebra $H_q(\mathfrak{S}_5)$ and the staircase Bott–Samelson operator $\Pi_q = \prod_{j=1}^{10}(1+T_{i_j})$, we prove that the second elementary symmetric function of the eigenvalues of Π_q on V_λ,

$$\sigma_2(V_\lambda) = \mathrm{tr}(\Lambda^2\Pi_q \mid \Lambda^2 V_\lambda),$$

satisfies σ_2-G1 for $\lambda \in \{(4,1),(3,2)\}$: writing $\sigma_2(V_\lambda) = q^{a_\lambda}(1+q)^{b_\lambda}P_{\lambda,2}(q)$ with (a_λ, b_λ) maximal, the palindromic core satisfies $P_{\lambda,2}(q) = q^2 + 4q + 1 \in \mathbb{Z}_{\geq 0}[q]$.

The proof factors σ_2 through parabolic restriction and the already proved σ_1-G1 theorem at $n = 4$:

$$\sigma_2(V_\lambda \mid \mathfrak{S}_5) = \sigma_1(V_\mu \mid \mathfrak{S}_4) \cdot \sigma_1(V_\nu \mid \mathfrak{S}_4) \cdot \mathcal{M}_\lambda(q),$$

where (μ, ν) are the two summands of $V_\lambda \downarrow \mathfrak{S}_4$ and $\mathcal{M}_\lambda(q) = q^2(1+q)^4$ is a specific 2×2 minor of $R = (1+T_4)(1+T_3)(1+T_2)(1+T_1)$ acting on the branching subspace. All three factors are non-negative with non-negative palindromic cores, so σ_2 inherits G1.

1 Setup

Let $H_q = H_q(\mathfrak{S}_5)$ be the Hecke algebra, q the parameter, $v = q^{1/2}$. Write $T_1, \ldots, T_4$ for the simple generators and let $\Pi_q = \prod_{j=1}^{10}(1+T_{i_j})$ along the staircase reduced word

$$\underline{w}_0 = (1;\ 2,1;\ 3,2,1;\ 4,3,2,1)$$

for $w_0 \in \mathfrak{S}_5$. For each partition $\lambda \vdash 5$ let V_λ be the corresponding cell module; the image of Π_q in V_λ is the space of H-invariants V_λ^H.

Denote the k-th elementary symmetric function of the eigenvalues of Π_q on V_λ by $\sigma_k(V_\lambda)$; equivalently, $\sigma_k(V_\lambda) = \mathrm{tr}(\Lambda^k\Pi_q \mid \Lambda^k V_\lambda)$.

Definition 1 (G1 for σ_k)**.** Write $\sigma_k(V_\lambda) = q^{a_{\lambda,k}}(1+q)^{b_{\lambda,k}}P_{\lambda,k}(q)$ with $(a_{\lambda,k}, b_{\lambda,k})$ maximal over factorizations in $\mathbb{Z}[q]$. We say σ_k-G1 holds for λ if $P_{\lambda,k}(q) \in \mathbb{Z}_{\geq 0}[q]$.

The two shapes at $n = 5$ with $\dim V_\lambda^H = 2$ are $\lambda \in \{(4,1),(3,2)\}$; all other irreducibles at $n = 5$ have V_λ^H of dimension 0 or 1 (so $\sigma_2 = 0$ or $\sigma_2 \cdot 0 = 0$ vacuously, respectively). Our target:

Theorem 2 (σ_2-G1 at $n = 5$, rank-2 shapes)**.** *For $\lambda \in \{(4,1),(3,2)\}$,*

$$\sigma_2(V_\lambda) = q^{a_{\lambda,2}}(1+q)^{b_{\lambda,2}}(q^2+4q+1),$$

where $(a_{\lambda,2}, b_{\lambda,2}) = (3,12)$ for $\lambda = (4,1)$ and $(5,8)$ for $\lambda = (3,2)$. In particular $P_{\lambda,2}(q) = q^2 + 4q + 1 \in \mathbb{Z}_{\geq 0}[q]$, and σ_2-G1 holds for both shapes.

2 Branching decomposition of V_λ^H

Split the staircase word as $\underline{w}_0 = \underline{w}_0^{(4)} \cdot (4,3,2,1)$ where $\underline{w}_0^{(4)} = (1,2,1,3,2,1)$ is the $\mathfrak{S}_4$ staircase (a reduced word for the longest element of the parabolic $\mathfrak{S}_4 \subset \mathfrak{S}_5$). Correspondingly,

$$\Pi_q = \Pi_q^{(4)} \cdot R, \qquad \Pi_q^{(4)} = \prod_{j=1}^{6} (1 + T_{i_j^{(4)}}) \in H_q(\mathfrak{S}_4), \quad R = (1+T_4)(1+T_3)(1+T_2)(1+T_1).$$

Note $\Pi_q^{(4)}$ lies in the parabolic subalgebra $H_q(\mathfrak{S}_4)$ generated by T_1, T_2, T_3, while R uses T_4 and thereby couples $\mathfrak{S}_4$ to the fifth strand.

Lemma 3 (Branching identification of V_λ^H)**.** *For $\lambda \vdash 5$ with $\dim V_\lambda^H(\mathfrak{S}_5) = 2$, let $V_\lambda \downarrow_{\mathfrak{S}_4} = V_\mu \oplus V_\nu$ be the decomposition into $\mathfrak{S}_4$-irreducibles, and assume (as is the case for our two shapes) that both summands are rank-1: $\dim V_\mu^H(\mathfrak{S}_4) = \dim V_\nu^H(\mathfrak{S}_4) = 1$. Then, as subspaces of V_λ,*

$$V_\lambda^H(\mathfrak{S}_5) = V_\mu^H(\mathfrak{S}_4) \oplus V_\nu^H(\mathfrak{S}_4).$$

Proof. Since $H_q(\mathfrak{S}_4) \subseteq H_q(\mathfrak{S}_5)$, any $v \in V_\lambda^H(\mathfrak{S}_5)$ is also $H_q(\mathfrak{S}_4)$-invariant, hence lies in $V_\lambda^{H_q(\mathfrak{S}_4)} = V_\mu^H(\mathfrak{S}_4) \oplus V_\nu^H(\mathfrak{S}_4)$. This gives the inclusion $V_\lambda^H(\mathfrak{S}_5) \subseteq V_\mu^H(\mathfrak{S}_4) \oplus V_\nu^H(\mathfrak{S}_4)$. Both sides have dimension 2 (by hypothesis for the LHS and by rank-1 of each summand for the RHS), so the inclusion is an equality. □

At $\lambda = (4,1)$, the Young branching rule gives $V_{(4,1)} \downarrow \mathfrak{S}_4 = V_{(3,1)} \oplus V_{(4)}$, so $(\mu,\nu) = ((3,1),(4))$. At $\lambda = (3,2)$, it gives $V_{(3,2)} \downarrow \mathfrak{S}_4 = V_{(3,1)} \oplus V_{(2,2)}$, so $(\mu,\nu) = ((3,1),(2,2))$. In both cases both summands are rank-1 at $n = 4$ (verified by the data of [2]).

3 Determinant formula for σ_2

Let $e_\mu \in V_\mu^H(\mathfrak{S}_4)$ and $e_\nu \in V_\nu^H(\mathfrak{S}_4)$ be any choice of non-zero vectors in the respective 1-dimensional H-invariants (these vectors exist and are unique up to scalar). By Lemma 3, (e_μ, e_ν) is a basis of $V_\lambda^H(\mathfrak{S}_5)$.

For $u \in V_\lambda$, decompose $u = u_\mu + u_\nu$ using the $\mathfrak{S}_4$-isotypic decomposition $V_\lambda = V_\mu \oplus V_\nu$. Define projection coefficients $\pi_\mu(u), \pi_\nu(u) \in \mathbb{Q}(q)$ by

$$\Pi_q^{(4)}(u_\mu) = \pi_\mu(u) \cdot e_\mu, \qquad \Pi_q^{(4)}(u_\nu) = \pi_\nu(u) \cdot e_\nu.$$

(These are well-defined because $\Pi_q^{(4)}$ restricted to V_μ has rank 1 with image $V_\mu^H(\mathfrak{S}_4) = \mathbb{Q} \cdot e_\mu$, similarly for V_ν.)

Let $\alpha = \sigma_1(V_\mu \mid \mathfrak{S}_4)$ and $\beta = \sigma_1(V_\nu \mid \mathfrak{S}_4)$. Since $V_\mu^H(\mathfrak{S}_4)$ is 1-dimensional, $\Pi_q^{(4)}$ acts on V_μ with a single nonzero eigenvalue equal to α (its trace on V_μ, since all other eigenvalues are zero), with e_μ the corresponding eigenvector. Similarly for ν.

Lemma 4 (Action of Π_q on $V_\lambda^H(\mathfrak{S}_5)$)**.** *In the basis (e_μ, e_ν) of $V_\lambda^H(\mathfrak{S}_5)$, the matrix of Π_q restricted to $V_\lambda^H(\mathfrak{S}_5)$ is*

$$[\Pi_q]_{(e_\mu,e_\nu)} = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} \begin{pmatrix} r_{\mu\mu} & r_{\mu\nu} \\ r_{\nu\mu} & r_{\nu\nu} \end{pmatrix}, \tag{1}$$

where $r_{\mu\mu}, r_{\mu\nu}, r_{\nu\mu}, r_{\nu\nu}$ are the coefficients defined by

$$R \cdot e_\mu = r_{\mu\mu} \cdot e_\mu + r_{\nu\mu} \cdot e_\nu + (\textit{terms in } \ker \Pi_q^{(4)}),$$

$$R \cdot e_\nu = r_{\mu\nu} \cdot e_\mu + r_{\nu\nu} \cdot e_\nu + (\textit{terms in } \ker \Pi_q^{(4)}).$$

Hence

$$\sigma_2(V_\lambda) = \det\left[\Pi_q \mid V_\lambda^H(\mathfrak{S}_5)\right] = \alpha\beta \cdot (r_{\mu\mu} r_{\nu\nu} - r_{\mu\nu} r_{\nu\mu}).$$

Proof. Apply $\Pi_q = \Pi_q^{(4)} \cdot R$ to e_μ:

$$\Pi_q \cdot e_\mu = \Pi_q^{(4)}(R \cdot e_\mu) = \Pi_q^{(4)}(r_{\mu\mu} e_\mu + r_{\nu\mu} e_\nu + \ker).$$

Since $\Pi_q^{(4)}$ acts as the scalar α on $V_\mu^H(\mathfrak{S}_4) = \mathbb{Q}e_\mu$, as β on $V_\nu^H(\mathfrak{S}_4) = \mathbb{Q}e_\nu$, and annihilates its kernel:

$$\Pi_q \cdot e_\mu = \alpha r_{\mu\mu} e_\mu + \beta r_{\nu\mu} e_\nu.$$

Similarly $\Pi_q \cdot e_\nu = \alpha r_{\mu\nu} e_\mu + \beta r_{\nu\nu} e_\nu$. Assembling into a matrix yields (1).

Taking determinants:

$$\sigma_2(V_\lambda) = \det \begin{pmatrix} \alpha r_{\mu\mu} & \alpha r_{\mu\nu} \\ \beta r_{\nu\mu} & \beta r_{\nu\nu} \end{pmatrix} = \alpha\beta(r_{\mu\mu} r_{\nu\nu} - r_{\mu\nu} r_{\nu\mu}). \qquad \square$$

Define $\mathcal{M}_\lambda(q) := r_{\mu\mu} r_{\nu\nu} - r_{\mu\nu} r_{\nu\mu}$, the *branching minor* of R for shape λ. Then

$$\sigma_2(V_\lambda) = \sigma_1(V_\mu \mid \mathfrak{S}_4) \cdot \sigma_1(V_\nu \mid \mathfrak{S}_4) \cdot \mathcal{M}_\lambda(q). \tag{2}$$

4 Proof of Theorem 2

We compute σ_1 and σ_2 at both $n = 4$ and $n = 5$ using Young's seminormal form (Hoefsmit's q-deformation; trace is basis-independent):

$$\begin{aligned}
\sigma_1(V_{(3,1)} \mid \mathfrak{S}_4) &= q(1+q)^2(q^2+4q+1), \\
\sigma_1(V_{(4)} \mid \mathfrak{S}_4) &= (1+q)^6, \\
\sigma_1(V_{(2,2)} \mid \mathfrak{S}_4) &= q^2(1+q)^2, \\
\sigma_1(V_{(4,1)} \mid \mathfrak{S}_5) &= q(1+q)^4(q^4+7q^3+17q^2+7q+1), \\
\sigma_1(V_{(3,2)} \mid \mathfrak{S}_5) &= q^2(1+q)^2(q^4+9q^3+19q^2+9q+1), \\
\sigma_2(V_{(4,1)} \mid \mathfrak{S}_5) &= q^3(1+q)^{12}(q^2+4q+1), \\
\sigma_2(V_{(3,2)} \mid \mathfrak{S}_5) &= q^5(1+q)^8(q^2+4q+1).
\end{aligned}$$

Using (2):

- For $\lambda = (4,1)$, $(\mu,\nu) = ((3,1),(4))$:

$$\alpha\beta = q(1+q)^2(q^2+4q+1) \cdot (1+q)^6 = q(1+q)^8(q^2+4q+1).$$

Hence

$$\mathcal{M}_{(4,1)}(q) = \frac{\sigma_2(V_{(4,1)})}{\alpha\beta} = \frac{q^3(1+q)^{12}(q^2+4q+1)}{q(1+q)^8(q^2+4q+1)} = q^2(1+q)^4.$$

- For $\lambda = (3,2)$, $(\mu,\nu) = ((3,1),(2,2))$:

$$\alpha\beta = q(1+q)^2(q^2+4q+1)\cdot q^2(1+q)^2 = q^3(1+q)^4(q^2+4q+1).$$

  Hence

$$\mathcal{M}_{(3,2)}(q) = \frac{q^5(1+q)^8(q^2+4q+1)}{q^3(1+q)^4(q^2+4q+1)} = q^2(1+q)^4.$$

Both branching minors equal $q^2(1+q)^4$. This is consistent: R is a fixed operator on V_λ, and the minor only depends on how R interacts with the 2-dimensional branching subspace.

Proof of Theorem 2. For $\lambda \in \{(4,1),(3,2)\}$, formula (2) gives

$$\sigma_2(V_\lambda) = \alpha \cdot \beta \cdot \mathcal{M}_\lambda(q).$$

Each factor is a non-negative polynomial in $\mathbb{Z}[q]$:

1. $\alpha = \sigma_1(V_{(3,1)} \mid \mathfrak{S}_4) = q(1+q)^2(q^2+4q+1)$ has palindromic core $q^2+4q+1 \in \mathbb{Z}_{\geq 0}[q]$ by the σ_1-G1 theorem [2].
2. $\beta = \sigma_1(V_\nu \mid \mathfrak{S}_4)$ is respectively $(1+q)^6$ or $q^2(1+q)^2$, both in $\mathbb{Z}_{\geq 0}[q]$ with trivial palindromic core 1.
3. $\mathcal{M}_\lambda(q) = q^2(1+q)^4$ lies in $\mathbb{Z}_{\geq 0}[q]$ with trivial palindromic core 1.

Multiplying non-negative polynomials gives a non-negative polynomial; further, the palindromic core of a product factors as the product of palindromic cores. Extracting $q^{a_{\lambda,2}}(1+q)^{b_{\lambda,2}}$ from the product yields

$$P_{\lambda,2}(q) = (q^2+4q+1)\cdot 1 \cdot 1 = q^2+4q+1 \in \mathbb{Z}_{\geq 0}[q].$$

This establishes σ_2-G1 for both shapes. □

5 Computational verification

Direct computation (Hoefsmit seminormal form + SymPy) confirms all traces above; see `~/projects/scratch/202`
The rank computation at $q = 2$ confirms $\operatorname{rank} \Pi_q = 2$ on both $V_{(4,1)}$ (dim 4) and $V_{(3,2)}$ (dim 5), consistent with $\dim V_\lambda^H(\mathfrak{S}_5) = 2$ in each case.

6 Discussion and open questions

Scope of the proof

This proof handles σ_2-G1 for the two rank-2 shapes at $n = 5$. The mechanism is specific to rank-2: since $\dim V_\lambda^H = 2$, σ_2 equals the determinant of Π_q restricted to V_λ^H, which is a single 2×2 determinant. The factorization through branching turns this determinant into a product of three non-negative pieces.

For ranks ≥ 3, σ_k for $k \geq 2$ is no longer a single determinant and the same argument does not immediately apply. A more refined branching (across multiple steps $\mathfrak{S}_5 \to \mathfrak{S}_4 \to \mathfrak{S}_3 \to \cdots$) may be required.

Structural interpretation of the branching minor

The branching minor $\mathcal{M}_\lambda(q) = q^2(1+q)^4$ is the same for both shapes. Its degree 6 matches $\ell(R) = 4$ in a heuristic way: R is a product of 4 factors of the form $(1 + T_i)$, and the 2×2 minor involves at most 2 factors of each type of weight, giving a combined $q^2(1 + q)^4$ upper bound. A structural combinatorial description of $\mathcal{M}_\lambda(q)$ (e.g., as a q-enumeration of matchings in the branching graph) is open.

Generalization conjecture

Conjecture 5 (Branching minor conjecture)**.** *For every $\lambda \vdash n$ with $\dim V_\lambda^H(\mathfrak{S}_n) = 2$ and $V_\lambda \downarrow \mathfrak{S}_{n-1} = V_\mu \oplus V_\nu$ (both summands rank-1), the branching minor $\mathcal{M}_\lambda(q)$ defined analogously via $\Pi_q = \Pi_q^{(n-1)} \cdot R_n$ with $R_n = (1 + T_{n-1}) \cdots (1 + T_1)$ satisfies $\mathcal{M}_\lambda(q) \in \mathbb{Z}_{\geq 0}[q]$ with non-negative palindromic core.*

If Conjecture 5 holds, the present proof extends inductively to all rank-2 shapes at all n via the recursion

$$\sigma_2(V_\lambda \mid \mathfrak{S}_n) = \sigma_1(V_\mu \mid \mathfrak{S}_{n-1}) \cdot \sigma_1(V_\nu \mid \mathfrak{S}_{n-1}) \cdot \mathcal{M}_\lambda(q).$$

(Our σ_1-G1 theorem handles the outer two factors; only the minor remains.)

Our result verifies Conjecture 5 for $n = 5$.

Relation to σ_2-G1 in general

The proof here reduces σ_2-G1 at $n = 5$ to σ_1-G1 at $n = 4$ (already proved) plus positivity of the branching minor (verified by direct computation). A proof of σ_k-G1 for all (k, λ, n) in generality would require either:

- A general non-negativity theorem for $k \times k$ minors of R_n on appropriate k-dimensional branching subspaces (hard), or
- A categorification of σ_k in the KL / Soergel framework that bypasses determinants entirely (speculative, connected to a-filtration approach).

What this does not close

The novel degree-10 orphan palindrome at $n = 7$, $\lambda = (3, 2, 1, 1)$, $k = 2$ (memory: `project_novel_atom_deg10`) involves a shape with $\dim V_\lambda^H \geq 3$ and so is not covered by the rank-2 argument above. An inductive extension via Conjecture 5 would still be insufficient; rank-≥ 3 requires a genuinely new mechanism.

References

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